

0 3 . 1

Below are **four** statements about the structure of prokaryotic cells.

1. No prokaryotic cell has DNA that is associated with proteins.
2. No prokaryotic cell has membrane-bound organelles.
3. All prokaryotic cells have one or more flagella.
4. All prokaryotic cells have smaller ribosomes than eukaryotic cells.

Which statements about the structure of prokaryotic cells are correct?

Tick (✓) **one** box.

[1 mark]

**A** statements **1, 2** and **3**

☐

**B** statements **1, 2** and **4**

☐

**C** statements **2, 3** and **4**

☐

**D** statements **1, 2, 3** and **4**

☐

A student investigated the effect of **two** antibiotics on the growth of the bacterium *Micrococcus luteus*.

During the investigation, the student:

- transferred 9 cm<sup>3</sup> of a liquid culture of *M. luteus* into each of **three** bottles
- added the antibiotic chloramphenicol to the first bottle
- added the antibiotic novobiocin to the second bottle
- added no antibiotic to the third bottle.

After 24 hours, he diluted the contents of each bottle by 1 in 100 000 ( $10^{-5}$ ). He then transferred 0.25 cm<sup>3</sup> samples from the first bottle onto each of 3 separate agar plates. He repeated this with 0.25 cm<sup>3</sup> samples from the second bottle and the third bottle, resulting in 9 agar plates in total.

He incubated the plates for 48 hours.

**Table 1** shows the number of colonies of bacteria he counted on each plate after 48 hours' incubation.

Table 1

| Plate | Number of colonies of bacteria |                    |                       |
|-------|--------------------------------|--------------------|-----------------------|
|       | with<br>chloramphenicol        | with<br>novobiocin | with<br>no antibiotic |
| 1     | 2                              | 238                | 276                   |
| 2     | 4                              | 263                | 258                   |
| 3     | 6                              | 261                | 324                   |

03.2

Calculate the mean number of bacteria in the **undiluted** bottle of liquid culture containing novobiocin.

Give your answer in standard form. Show your working.

[2 marks]

Answer \_\_\_\_\_ bacteria

03.3

Starting with a single bacterium, calculate how many generations it would take to produce at least the number of bacteria you have calculated for your answer in Question 03.2

You can assume no bacteria die.

You could use the ln or log button on your calculator to calculate your answer.

[1 mark]

Answer \_\_\_\_\_ generations

Question 3 continues on the next page

0 3 . 4

*M. luteus* is **not** resistant to chloramphenicol.

Suggest **two** reasons why the bacteria were able to grow in the culture containing chloramphenicol.

[2 marks]

1 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

[  
-  
]

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ANSWER IN THE SPACES PROVIDED**

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